

Omkar Jois

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ACADEMIC QUALIFICATION

- **Bachelor of Technology in Computer Science (AI & ML)** | PES University, Bangalore | 2021-2025 | **CGPA: 9.09/10.00**
 - Awarded **40% Merit Scholarship** for consistent performance in the **top 20% of the Computer Science Department**, achieved in Semesters 2, 4, 5, and 6.

EXPERIENCE

Core Engineering Team Member, Kshana.ai., Bangalore

Aug 2025 –

Present

(Kshana.ai is an AI co-pilot that transforms financial data into actionable insights for asset managers.)

- Designed and implemented a **data platform** that collects and preprocesses financial data (using Python, Docker, and PostgreSQL) to power the company's AI engine for stock market insights.
- Automated data ingestion and scheduling with **Apache Airflow** and **Azure Blob Storage**, improving task efficiency and scalability.
- Built agent-based data extraction workflows with **LangChain** and **Playwright**, enabling structured processing of raw data from public sources such as NSE, BSE, and SEC filings.
- Reduced **data acquisition and preprocessing time** from ~1 day to **under 1 hour**, accelerating insights for financial analysts.
- Collaborated closely with the founding team to align data collection pipelines with model requirements, while working independently on implementation.
- Contributed to a **production-level system** deployed within the product, as part of a **startup team of 8** (3 co-founders, 3 engineers, 2 business/marketing).

Co-Founder, PocketCoach Technologies Pvt. Ltd., Bangalore

Jan 2023 –

Present

(PocketCoach is an AI-powered sports training app that gamifies drills, tracks athlete performance, and provides data-driven insights for coaches and players. Founded by a team of 3 students from PES University.)

- Secured **\$70,000+ in funding** through the Startup India Seed Fund (via IIMB), Startup Karnataka Elevate Program (2025), and won **1st place nationally** in Cisco thingQbator Cohort 5 (\$6,000) among 1,000+ teams.
- Selected for the **IIMB NSRCEL Campus Founders Program**, gaining mentorship and entrepreneurial exposure.
- Built a **real-time video analytics pipeline** (Python, AWS EC2) for gameplay streaming, later migrated to **on-device AI modules** on iOS to cut latency and reduce server costs drastically.
- Developed a **YOLO-based basketball analysis system** (9,000+ annotated images) with ~90% shot-tracking accuracy, plus **gamified training drills** using pose estimation and custom detection models.
- Led development of the **PocketCoach mobile app** (Flutter, with Swift/Java integrations), expanding the gamified exercise library from 25 to 75+ drills and designing the app's UI/UX.
- Partnered with **One World International School** and **Tenvic Sports Academies**, converting pilots into paid subscriptions and driving adoption through a **B2B2C business model**.
- Grew the platform to **1,500+ downloads** across Android/iOS with ~200 active users, building traction in the youth sports training ecosystem.
- Managed and mentored a **team of 3 interns** to accelerate feature development and expand the drill offerings, while also contributing to investor pitches and product strategy.

Summer Research Intern, CDSAML

Jun 2023 – Aug

2023

(Center for Data Science and Applied Machine Learning), PES University, Bangalore

(A research-focused center fostering innovation in data science and machine learning through faculty-led projects and industry collaboration.)

- Designed and implemented **deep learning models (LSTMs, Transformers)** to automatically classify AWS CloudWatch logs into **CWE (Common Weakness Enumeration) error categories**, improving security monitoring and threat detection.
- Built **NLP pipelines** to process unstructured log data and detect patterns linked to vulnerabilities.
- Developed a **CVSS score estimator** to automatically rank vulnerabilities by severity, enabling faster risk prioritization.
- Achieved **95% classification accuracy** on cloud log datasets, establishing one of the first works in this space.
- Conducted research in collaboration with **CDSAML (under Dr. Sujata)** and **SkureCloud**, a cloud security startup, as part of R&D for their product.
- Published the results in a peer-reviewed research paper: [Logs2Vul: Vulnerability Detection from Logs for CSPM](#)

Software Development Intern, Hajdro, Bangalore

Sept 2021 – Jan

2022

(Hajdro develops automated hydroponic systems with Wi-Fi enabled monitoring for real-time control and optimization of plant growth.)

- Programmed the **ESP-32 microcontroller in C++**, implementing **Wi-Fi connectivity** and **Over-the-Air (OTA) firmware updates** for seamless remote upgrades.
- Designed and optimized **embedded system workflows** for reliable device communication and performance during the R&D phase.
- Independently developed and tested connectivity features, enabling hydroponic systems to be monitored and controlled remotely over Wi-Fi.
- Built and enhanced **web interface features** using **React.js, HTML, and CSS**, allowing users to monitor real-time system updates.

COMPETITIONS & AWARDS

- **IIM Bangalore NSRCEL Campus Founders Program (Nov 2024 – Mar 2025)**: Chosen as one of **India's top student-led startups**; gained structured mentorship, workshops, and investor access to scale PocketCoach.
- **Startup India Seed Fund Awardee (Jan 2025)**: Received **\$35,000** funding through **IIM Bangalore NSRCEL** to advance PocketCoach's R&D and growth.
- **ELEVATE Idea2PoC (Oct 2024)**: Selected from **700+** startups for a **\$35,000** grant by the Govt. of Karnataka, recognizing PocketCoach's innovation in sports technology.
- **Kodikon 24-hour Hackathon (Nov 2023)**: Secured **1st place** for developing a full-stack sports tournament management app enabling automated registrations, scheduling, payments, and notifications.
- **Cisco thingQbator Cohort 5 Winner (Apr 2022 – Feb 2023)**: Ranked **1st nationally** in Cisco & NASSCOM Foundation's innovation program, receiving **\$6,000** to advance PocketCoach.

INDEPENDENT PROJECTS & RESEARCH

Topic: Computer Vision ([Link to Publication](#))

Project Name: Positional analysis using Monocular Relative Depth and Angular Estimation

Duration: Jun 2024 - Sept 2024

Tools Used: Python

Summary: Developed a **monocular vision algorithm** that combines relative depth estimation with angular analysis and a calibration step to achieve **higher positional accuracy than MiDaS**, enabling cost-effective player tracking for sports analytics. Built primarily with **OpenCV** and **NumPy**, the method was tested on numerous basketball courts showing strong adaptability to change in environments. The approach was later **integrated into the shooting feature of PocketCoach** and was published as an IEEE conference paper in the proceedings of the 2025 **14th International Conference on Information Technology in Asia (CITA)**.

Topic: Generative AI ([GitHub](#))

Project Name: Enhancing Object Detection through Generative AI-Based Data Augmentation Techniques

Duration: Mar 2024 - Apr 2024

Tools Used: Python

Summary: Explored and implemented **generative AI-based data augmentation** techniques to improve object detection on limited and imbalanced datasets, using **Stable Diffusion** to preserve target objects while varying contextual details. The approach was benchmarked on **YOLOv5** with two datasets (a 300-image custom bottle dataset and a multi-class vehicle dataset), achieving an **average accuracy improvement of 9%** compared to traditional augmentation methods. The project was conducted under the guidance of **Dr. Uma (Deep Learning Professor)** and later adapted for use in **PocketCoach's basketball detection model**, demonstrating practical applicability.

CERTIFICATIONS

- **Neural Networks and Deep Learning** | Coursera | *Aug 2023*
- **App Development with Flutter** | PESU IO | *Oct 2021*

EXTRACURRICULARS & LEADERSHIP

- **Wildlife Photographer & Trekker (2021 – Present)**: Photographed Indian wildlife and completed 10+ Himalayan treks; work recognized in the 28th INW Awards collection.
- **Head – Pixels Photography Club (Sept 2023 – Sept 2024)**: Managed a 40+ member team, curated photography initiatives, and promoted campus culture through events and visual media.
- **Core Team, Aatmatrisha Fest (Jan 2024 – Apr 2024)**: Managed photography operations with a 30+ member team, providing comprehensive coverage of 100+ cultural and concert events.
- **College & Club Football (2019 – 2022)**: Consistently represented PES University and BJFC in major tournaments; winner of MFAR-BOCA Cup (2019), finalist at CBSE Clusters (2020), and runner-up at BMS Kreedotsav (2021).
- **Karate-Do (2008 – 2016)**: Trained under the Okinawa Shorin Ryu Shorin Kan Karate-Do Federation of India; achieved Black Belt certification in 2015.